

Topic Modeling and Link-Prediction for Material Property Discovery

Ryan C. Barron
barron@lanl.gov
Theoretical Division,
Los Alamos National Laboratory
Los Alamos, New Mexico, USA
CSEE, University of Maryland,
Baltimore County
Baltimore, Maryland, USA

Maksim E. Eren
Information Systems and Modeling,
Los Alamos National Laboratory
Los Alamos, New Mexico, USA

Valentin Stanev
Department of Material Science &
Engineering, University of Maryland
College Park, Maryland, USA

Cynthia Matuszek
CSEE, University of Maryland,
Baltimore County
Baltimore, Maryland, USA

Boian S. Alexandrov
Theoretical Division,
Los Alamos National Laboratory
Los Alamos, New Mexico, USA

ABSTRACT

Link prediction is a key network analysis technique that infers missing or future relations between nodes in a graph, based on observed patterns of connectivity. Scientific literature networks and knowledge graphs are typically large, sparse, and noisy, and often contain missing links, potential but unobserved connections, between concepts, entities, or methods. Here, we present an AI-driven hierarchical link prediction framework that integrates matrix factorization to infer hidden associations and steer discovery in complex material domains. Our method combines Hierarchical Nonnegative Matrix Factorization (HNMfK), Boolean matrix factorization (BNMfK) with automatic model selection. These discrete factors are then fused with Logistic matrix factorization (LMF), we use to construct a three-level topic tree from a 46,862-document corpus focused on 73 transition-metal dichalcogenides (TMDs). This class of materials has been studied in a variety of physics fields and has a multitude of current and potential applications.

An ensemble BNMfK + LMF approach fuses discrete interpretability with probabilistic scoring. The resulting HNMfK clusters map each material onto coherent research themes, such as superconductivity, energy storage, and tribology, and highlight missing or weakly connected links between topics and materials, suggesting novel hypotheses for cross-disciplinary exploration. We validate our method by removing publications about superconductivity in well-known superconductors, and demonstrate that the model correctly predicts their association with the superconducting TMD clusters. This highlights the ability of the method to find hidden connections in a graph of material to latent topic associations built from scientific literature. This is especially useful when examining a diverse corpus of scientific documents covering the same class of phenomena or materials but originating from distinct communities and

perspectives. The inferred links generating new hypotheses, produced by our method, are exposed through an interactive Streamlit dashboard, designed for scientific discovery.

CCS CONCEPTS

• **Applied computing** → **Chemistry**; • **Computing methodologies** → **Probabilistic reasoning**.

KEYWORDS

NMF, NMfK, LMF, Link Prediction, Matrix completion

ACM Reference Format:

Ryan C. Barron, Maksim E. Eren, Valentin Stanev, Cynthia Matuszek, and Boian S. Alexandrov. 2025. Topic Modeling and Link-Prediction for Material Property Discovery. In *ACM Symposium on Document Engineering 2025 (DocEng '25)*, September 2–5, 2025, Nottingham, United Kingdom. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3704268.3748680>

1 INTRODUCTION

Artificial intelligence is transforming materials science by addressing key challenges in discovery, synthesis, and property prediction. Recent advances, particularly in large language models (LLMs), transformers, and hybrid systems, have demonstrated strong performance in overcoming traditional bottlenecks [10, 17].

Here we present a novel, model-agnostic framework for missing link prediction between latent features extracted from a domain-specific ontology of 46,862 scientific papers on the 73 known transition metal dichalcogenides (TMDs) [2]. The proposed workflow integrates: a) the BUNIE method [15] to build a targeted TMDs ontology, b) HNMfK [7] with automatic topic number selection and interactive visualization tool, to extract latent topics, and c) a custom Boolean Matrix Factorization with model selection, combined with Logistic Matrix Factorization (BNMfK-LMF) [1], to identify missing links in the **Materials Property Matrix**, derived from the TMDs corpus and HNMfK decomposition¹.



¹NMfK, HNMfK, BNMfK, and LMF are available at <https://github.com/lanl/T-ELF> [6].

Our approach constructs a binary **Materials Property Matrix** from the extracted topics and material references, enabling prediction of functional links. We show that the proposed approach can effectively distinguish true superconductors from chemically similar non-superconductors and prioritize candidates for experimental validation. Its generalizable architecture supports broad application across material families and incomplete relational datasets, accelerating data-driven scientific discovery.

2 RELATED WORK

Matrix completion algorithms have been specifically developed for discrete rating-scale data, addressing challenges such as corrupted observations and missing entries commonly encountered in recommender systems, surveys, and other partially observed datasets [3]. Missing link prediction and matrix completion are closely related tasks, as both aim to recover missing values in incomplete matrices or graphs by exploiting underlying structure [5, 13]. In the context of link prediction, the adjacency matrix of a graph is typically assumed to be low-rank, enabling the use of matrix completion techniques, such as low-rank approximation or factorization, to estimate the likelihood of unobserved connections [9]. Classical non-negative matrix factorization (NMF)-based link scoring continues to serve as a basis approach in this domain [4].

For binary (categorical) networks with $\{0, 1\}$ edge structures, missing link prediction seeks to infer unobserved relationships based on observed binary interactions. Logistic Matrix Factorization (LMF) models the probability of a link by applying a sigmoid transformation to the dot product of latent embeddings [11, 12], whereas Boolean Matrix Factorization (BMF) directly approximates the binary adjacency matrix using logical operations such as AND and OR, preserving interpretability in discrete domains [14]. For predicting material–property associations, [16] use a hypergraph of articles combining entities and author collaborations, using embeddings and human-pathway-based transition probabilities. In contrast, our approach performs link prediction over latent topic–material associations to infer material–property relationships.

3 METHODS

3.1 Corpus Curation with BUNIE

We begin with the 46,862-document TMD literature. The interactive tool BUNIE [15] allows a domain expert to iteratively filter this collection by relevance, producing a high-precision subset focused on the 73 crystallographically confirmed TMD compounds. Compared with keyword filtering, BUNIE removes off-topic or tangential papers for cleaner input to topic modeling.

3.2 Materials–Properties Matrix

From the curated corpus we extract:

- (1) a TF-IDF document-term matrix and
- (2) a list of material mentions per document (exact string match to the 73 compounds).

Applying HNMfK (next subsection) to the TF-IDF matrix produces T latent topics. We therefore construct a Materials–Properties Matrix $\mathbf{X} \in \{0, 1, \text{NaN}\}^{T \times 73}$,

$$X_{ij} = \begin{cases} 1 & \text{if topic } t_i \text{ co-occurs with material } m_j, \\ 0 & \text{if the co-occurrence is } \textit{observed} \text{ to be absent,} \\ \text{NaN} & \text{if the entry is unobserved/unknown.} \end{cases}$$

Throughout the factorization steps, a binary mask $\mathbf{1}_{[X_{ij} \neq \text{NaN}]}$ excludes unknown entries from the loss, while allowing the final model to score them.

3.3 Hierarchical Non-negative Matrix Factorization (HNMfK)

Starting from the full TF-IDF matrix A and an index set $I^{(0)} = \{1, \dots, n\}$, HNMfK recursively applies NMFk to ever smaller submatrices:

- (1) **Local factorisation.** Build $A^{(d)}$ from $I^{(d)}$, predict the stable rank k_d via NMFk, and factorise $A^{(d)} \approx W^{(d)}H^{(d)}$.
- (2) **Cluster assignment.** Assign each sample $i \in I^{(d)}$ to $\ell_i = \arg \max_r W_{ir}^{(d)}$ (or $H^{(d)}$), yielding k_d clusters.
- (3) **Stopping test.** Halt if any condition holds: $\text{mindim } A^{(d)} \leq 1$, $|I^{(d)}| \leq s_{\min}$, $k_d = 1$, all ℓ_i identical, or depth $d = d_{\max}$.
- (4) **Recursion.** For each non-empty cluster c set $I_c^{(d+1)} = \{i \in I^{(d)} \mid \ell_i = c\}$, choose

$$K^{(d+1)} = \{k_{\min}, \dots, \min(k_{\max}, k_d + 1)\},$$

and repeat steps 1–4 on $A_c^{(d+1)}$.

Adaptive rank selection and automatic stopping yield a multiscale topic hierarchy without manual tuning. The leaf topics define the row set of $\mathbf{X}$ in Sec. 3.2.

3.4 Logistic Matrix Factorization (LMF)

LMF extends low-rank factorization to binary data. Given $\mathbf{X}$ and its mask $\mathbf{M}$,

$$\hat{\mathbf{X}} = \sigma(\mathbf{WH} + \mathbf{b}_r + \mathbf{b}_c), \quad \sigma(x) = \frac{1}{1 + e^{-x}},$$

where $\mathbf{W} \in \mathbb{R}^{T \times k}$, $\mathbf{H} \in \mathbb{R}^{k \times 73}$, and $\mathbf{b}_r, \mathbf{b}_c$ are row/column biases. The regularized negative log-likelihood is

$$\mathcal{L} = -\sum_{i,j} M_{ij} [X_{ij} \log \hat{X}_{ij} + (1 - X_{ij}) \log(1 - \hat{X}_{ij})] + \lambda(\|\mathbf{W}\|_F^2 + \|\mathbf{H}\|_F^2 + \|\mathbf{b}_r\|_2^2 + \|\mathbf{b}_c\|_2^2), \quad (1)$$

with gradients computed only on observed entries ($M_{ij} = 1$). The logistic link returns probabilities for downstream ranking.

3.5 BNMFk–LMF Ensemble

BNMFk is the Boolean counterpart of NMFk, enforcing $\{0, 1\}$ constraints on the factors while selecting the rank k via the NMFk stability heuristic. To combine BNMFk’s interpretability with LMF’s calibrated scores we employ the ensemble of [1]:

- (1) **Boolean stage.** Apply BNMFk to $\mathbf{X}$, obtaining $\hat{\mathbf{X}} = \mathbf{WH} \in \{0, 1\}^{T \times 73}$.
- (2) **Logistic stage.** Holding k fixed, fit an LMF model to learn biases $\mathbf{b}_r, \mathbf{b}_c$ on the same $\mathbf{X}$.

(3) **Ensembling.**

$$\tilde{X} = \sigma(\hat{X} + b_r + b_c),$$

which assigns a probability to every (possibly NaN) entry.

This ensemble preserves the discrete structure uncovered by BNMFk while providing well-calibrated link probabilities for the unobserved material–topic pairs.

4 RESULTS

4.1 Materials–Property Matrix

The curated TMD corpus comprises 46,862 peer-reviewed articles. TMDs have the general formula MX_2 (M = transition metal, X = chalcogen) and underpin much of 2-D materials research because of their outstanding electronic, optical, and mechanical behavior [8]. Although only 73 compounds are experimentally verified, the combinatorial MX_2 design space is vast and largely unexplored.

Topic extraction. Titles and abstracts selected with BUNIE are vectorized with TF-IDF and decomposed by HNMF_k. A depth of **three** levels ($L = 3$) maximizes bootstrap stability, giving coarse “super-topics” at the root and progressively finer sub-topics below.

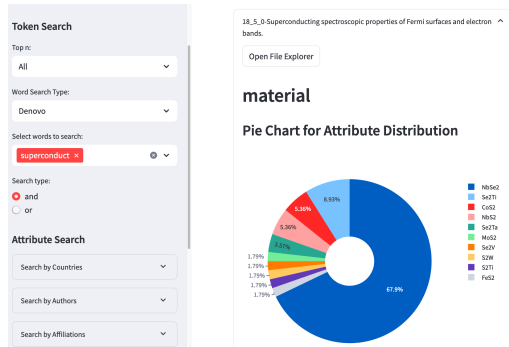


Figure 1: Minimal dashboard for inspecting HNMFk topics and their material distributions.

Interactive validation. A lightweight Streamlit dashboard lets users (i) browse the HNMF_k tree, (ii) filter by keywords or metadata, and (iii) click through from any cluster to the underlying documents (Fig. 1). This tight human loop was used only for qualitative checks and does not influence the automated link-prediction pipeline.

Binary matrix. Combining the $T = 815$ leaf topics with the 73 materials yields a $T \times 73$ Materials–Property Matrix M , where $M_{ij} = 1$ if topic t_i co-occurs with material m_j , $M_{ij} = 0$ if an observed absence is recorded, and $M_{ij} = \text{NaN}$ if the relationship is unknown.

4.2 Top k retrieval of hidden links

To evaluate whether our BNMFk + LMF ensemble can rediscover true but withheld superconducting links, we perform a leave-out experiment on four benchmark transition-metal dichalcogenides: NbSe₂, MoS₂, S₂Ta, and Se₂Ta.

- (1) **Masking.** For each compound, remove all verified superconducting links from the compound–superconductor cluster matrix, and randomly sample an equal number of non-superconducting (zero) entries as negatives.

- (2) **Training and scoring.** Train the ensemble on the masked matrix and compute predicted scores s_{ij} for each masked entry (i, j) .

- (3) **Metric.** Define

$$\text{hit}@k = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \mathbf{1}\{\text{rank}(s_{ij}) \leq k\},$$

where $\mathcal{P}$ is the set of masked positives, and $\text{rank}(s_{ij})$ is the score’s position among all masked entries sorted in descending order.

- (4) **Cross-validation.** Repeat the above over three random splits to estimate 95% confidence intervals.

Figure 3 shows mean hit@1 and hit@3 across folds. Key findings:

- hit@1 > 0.885 for all compounds,
- hit@1 = 1.000 for MoS₂,
- hit@3 = 1.000 for all four.

The narrow 95% intervals confirm high consistency across folds. These results demonstrate that our ensemble reliably rediscovers masked superconducting links with near-perfect top-3 precision.

4.3 Separation of positive and negative classes

Figure 2 contrasts the posterior scores assigned to the 24 withheld positive edges (green violins) with the 192 masked negatives (blue). Positive edges cluster tightly near 1: the median sits above 0.90 and the lower quartile rarely drops below 0.85. Negatives are strongly skewed toward zero (median < 0.05); fewer of the 192 instances per material penetrate the upper quartile. The cumulative counts printed on each quartile boundary emphasize the clean separation: most positives occupy the top quartile, whereas the vast majority of negatives remain in the lowest.

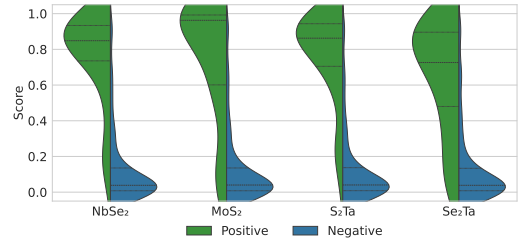


Figure 2: Split-violin plot of posterior probabilities for 24 superconducting edges (green). [See Table 1.] 192 non-superconducting edges (blue) were added for comparison (identical for the four violins.) Lines denote the median and 25th/75th percentiles, with numbers showing cumulative counts. Superconductors concentrate near 1, non-superconductors near 0, indicating a clear separation.

4.4 Ranking of unseen candidate materials

Table 1 reports the posterior score assigned to each of the eleven compounds whose labels were fully masked during training. In other words, every entry that would have indicated “compound i is in a superconductor cluster” was removed from the observation matrix (masked as unknown), so the model never saw a positive label

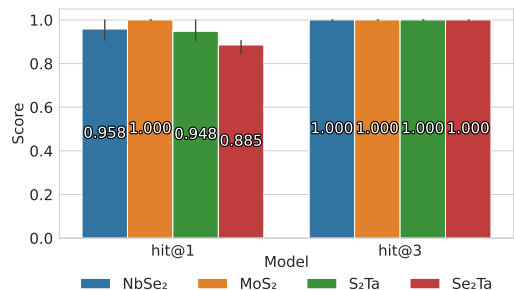


Figure 3: Mean hit@1 and hit@3 scores (bars, annotated) of the BNMf + LMF ensemble after masking every known superconducting link and an equal number of random zeros for each of the four benchmark compounds

Table 1: Average predicted scores for selected materials

Material	Set	Prediction Score
S ₂ Ta	Superconductor	0.813
NbSe ₂	Superconductor	0.757
MoS ₂	Superconductor	0.703
Se ₂ Ta	Superconductor	0.699
FeS ₂	Non-superconductor	0.206
MnS ₂	Non-superconductor	0.116
CrS ₂	Non-superconductor	0.109
MnSe ₂	Non-superconductor	0.105
MnTe ₂	Non-superconductor	0.094
CrSe ₂	Non-superconductor	0.079
CrTe ₂	Non-superconductor	0.052

for these materials while it was being fit. The corresponding row and column themselves were retained, populated only with zeros or missing values, so the model could still exploit their similarity to the rest of the data.

Importantly, the four layered dichalcogenides that are experimentally known to superconduct (S_2Ta , $NbSe_2$, MoS_2 , Se_2Ta) emerge at the very top of the ranking with scores in the range 0.70–0.81, whereas every material with no reported superconductivity receives a score ≤ 0.206 . The wide gap between the two groups implies that a single global threshold (≈ 0.5) would recover all known positives while rejecting all inspected negatives, demonstrating that the model can correctly infer superconducting behavior even when the ground-truth labels are hidden during training.

Taken together, the high hit@ k retrieval, the clear score separation, and the compound-level ranking show that the proposed approach: (i) rediscovers hidden superconducting links with essentially perfect precision and (ii) assigns suitably low confidence to chemically similar non-superconducting materials, highlighting its utility for guiding targeted experimental exploration.

5 CONCLUSION

Using BUNIE, HNMf, and BNMf+LMF, we analyzed a corpus spanning 73 known TMDs and built a sparse binary Material–Property Matrix (materials vs. hidden topics). We masked known superconducting and non-superconducting TMDs from the matrix and tested recovery without access to the withheld labels. Our model:

- recovered all masked superconductors within the top three predictions (hit@3 = 1.00), with 90 % ranked first,
- assigned high scores to positives (median ≈ 0.91) and low scores to most negatives (< 0.20), and

- ranked all four superconductors above all non-superconductors in a 15-compound test set, with a >0.5 score gap.

This model-agnostic workflow reliably distinguishes true superconductors from similar compounds and is transferable to other material families or incomplete relational datasets.

ACKNOWLEDGMENTS

This research was funded by the U.S. Department of Energy National Nuclear Security Administration’s Office of Defense Nuclear Nonproliferation Research and Development (DNN R&D), supported by the U.S. DOE NNSA under Contract No. 89233218CNA000001, as well as by the LANL Institutional Computing Program.

REFERENCES

- [1] Ryan Barron, Maksim E Eren, Duc P Truong, Cynthia Matuszek, James Wendelberger, Mary F Dorn, and Boian Alexandrov. 2025. Matrix Factorization for Inferring Associations and Missing Links. *arXiv preprint arXiv:2503.04680* (2025).
- [2] Manish Bhattarai, Ryan Barron, Maksim Eren, Minh Vu, Vesselin Grantcharov, Ismael Boureima, Valentin Stanev, Cynthia Matuszek, Vladimir Valtchinov, Kim Rasmussen, and Boian Alexandrov. 2024. HEAL: Hierarchical Embedding Alignment Loss for Improved Retrieval and Representation Learning. *arXiv:2412.04661* [cs.LG]. <https://arxiv.org/abs/2412.04661>
- [3] Emmanuel J Candes and Yaniv Plan. 2010. Matrix completion with noise. *Proc. IEEE* 98, 6 (2010), 925–936.
- [4] Bolun Chen, Fenfen Li, Senbo Chen, Ronglin Hu, and Ling Chen. 2017. Link prediction based on non-negative matrix factorization. *PLOS ONE* 12, 8 (2017), e0182968. <https://doi.org/10.1371/journal.pone.0182968>
- [5] Daniel M Dunlavy, Tamara G Kolda, and Evrim Acar. 2011. Temporal link prediction using matrix and tensor factorizations. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 5, 2 (2011), 1–27.
- [6] Maksim Eren, Nick Solovyyev, Ryan Barron, Manish Bhattarai, Duc Truong, Ismael Boureima, Erik Skau, Kim Rasmussen, and Boian Alexandrov. 2023. *Tensor Extraction of Latent Features (T-ELF)*. Technical Report. Los Alamos National Laboratories. <https://doi.org/10.5281/zenodo.10257897>
- [7] Maksim E Eren, Manish Bhattarai, Robert J Joyce, Edward Raff, Charles Nicholas, and Boian S Alexandrov. 2023. Semi-supervised classification of malware families under extreme class imbalance via hierarchical non-negative matrix factorization with automatic model selection. *ACM Transactions on Privacy and Security* 26, 4 (2023), 1–27.
- [8] Sunny Gupta, Jun-Jie Zhang, Jincheng Lei, Henry Yu, Mingjie Liu, Xiaolong Zou, and Boris I Yakobson. 2025. Two-Dimensional Transition Metal Dichalcogenides: A Theory and Simulation Perspective. *Chemical Reviews* (2025).
- [9] Seyed Mehran Kazemi and David Poole. 2018. Simple embedding for link prediction in knowledge graphs. *Advances in neural information processing systems* 31 (2018).
- [10] Siyu Liu, Tongqi Wen, Beilin Ye, Zhuoyuan Li, Han Liu, Yang Ren, and D Srolovitz. 2025. Large Language Models for Material Property Predictions: elastic constant tensor prediction and materials design. *Digital Discovery* (2025).
- [11] Yong Liu, Min Wu, Chunyan Miao, Peilin Zhao, and Xiao-Li Li. 2016. Neighborhood regularized logistic matrix factorization for drug-target interaction prediction. *PLoS computational biology* 12, 2 (2016), e1004760.
- [12] Aditya Krishna Menon and Charles Elkan. 2011. Link prediction via matrix factorization. In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2011, Athens, Greece, September 5–9, 2011, Proceedings, Part II* 22. Springer, 437–452.
- [13] Ratha Pech, Dong Hao, Liming Pan, Hong Cheng, and Tao Zhou. 2017. Link prediction via matrix completion. *Europhysics Letters* 117, 3 (2017), 38002.
- [14] Siamak Ravanbakhsh, Barnabás Póczos, and Russell Greiner. 2016. Boolean matrix factorization and noisy completion via message passing. In *International Conference on Machine Learning*. PMLR, 945–954.
- [15] Nicholas Solovyyev, Ryan Barron, Manish Bhattarai, Maksim E Eren, Kim Ø Rasmussen, and Boian S Alexandrov. 2023. Interactive distillation of large single-topic corpora of scientific papers. In *2023 International Conference on Machine Learning and Applications (ICMLA)*. IEEE, 1000–1005.
- [16] Jamshid Sourati and James A Evans. 2023. Accelerating science with human-aware artificial intelligence. *Nature human behaviour* 7, 10 (2023), 1682–1696.
- [17] Joren Van Herck, Maria Victoria Gil, Kevin Maik Jablonka, Alex Abrudan, Andy S Anker, Mehrdad Asgari, Ben Blaiszik, Antonio Buffo, Leander Choudhury, Clemence Corminboeuf, et al. 2025. Assessment of fine-tuned large language models for real-world chemistry and material science applications. *Chemical science* 16, 2 (2025), 670–684.